

SCHOOL OF COMPUTING AND INFORMATICS

Course Code	CCE2233	Course Name	Requirements Engineering
Lecturer	Mdm. Nadiah Arsat		
Semester	2, 2025/2026		
Assessment	Individual Assignment		
Weightage (%)	20		
Marks	50		
Course Learning Outcome	CLO2 - Describe software requirement engineering processes using appropriate tools and techniques (C2, PLO6)		

Lab Group	BCS3B-C2
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Total Marks (50)	

Assignment Rubrics

No	Criteria	Weight	Exemplary (4 points)	Proficient (3 points)	Developing (2 points)	Beginning (1 point)	
1	Requirements Elicitation and Documentation	10 M	Clearly defines ≥8 requirements (≥5 FR, ≥3 NFR) with complete, accurate, and well-structured descriptions	Requirements mostly complete with minor issues in clarity or structure	Some requirements missing or unclear	Incomplete, incorrect, or poorly defined requirements	
2	Modelling (Use Case)	10 M	Use case diagram and descriptions are accurate, complete, and fully aligned with requirements	Minor errors in diagram or descriptions; mostly aligned	Basic diagram with missing elements or weak descriptions	Incorrect or missing use case diagram/descriptions	
3	Requirements Prioritisation (MoSCoW)	6 M	Correct and justified prioritisation using MoSCoW; clear rationale	Mostly correct prioritisation with limited justification	Basic prioritisation with weak or unclear reasoning	Incorrect or missing prioritisation	
4	Traceability Matrix	6 M	Complete and accurate mapping between requirements and use cases	Minor gaps or inconsistencies in mapping	Partial mapping; some missing links	Missing or incorrect traceability matrix	
5	Tool Utilization	6 M	Uses 5 or more appropriate tools effectively; strong justification and clear evidence (screenshots/output)	Uses 3-4 tools appropriately with some justification and evidence	Uses 1-2 tools with limited explanation	No tool used or unclear usage	
6	Report Structure & Organisation	6 M	Fully follows required structure; clear, professional, and well-organised	Minor issues in structure or clarity	Some sections missing or poorly organised	Poor structure; many missing sections	
7	Reflection	3 M	Insightful reflection on challenges and clear explanation of how tools/techniques addressed them	Adequate reflection with some insights	Basic reflection with limited depth	Minimal or no reflection	
8	References (APA Style)	3 M	≥3 credible sources; correct APA format with no errors	Meets minimum sources with minor APA errors	Fewer sources or inconsistent APA style	Missing or incorrect referencing	

Table of Contents

1. Introduction	3
2. Stakeholder Analysis	3
3. Requirements Elicitation and Documentation	4
3.1 Functional Requirements	4
3.2 Non-Functional Requirements	5
4. Use Case Modelling	5
4.1 Use Case Diagram	6
4.2 Use Case Descriptions	7
5. Requirements Prioritisation (MoSCoW Technique)	9
6. Traceability Matrix	10
7. Tool Usage Explanation	11
7.1 Draw.io	11
7.2 Wiki Technologies.....	12
7.3 Trello.....	12
7.4 GitHub	13
7.5 ReqView	13
8. Conclusion.....	14
9. Reflection	14
10. References.....	15

1. Introduction

2. Stakeholder Analysis

Stakeholder analysis is of importance in Requirements Engineering as it can be used to determine who the people or groups are that interact with the system and will be impacted by its functions. By identifying stakeholders, the requirements of the system can be accurately collected from user needs and expectations. AIU Moodle LMS provides interaction between various stakeholders for academic, administrative, and technical purpose. If they understand their role and interests, then the system is able to support effective teaching, learning and system management activities.

Stakeholder	Role in the System	Needs / Interests
Students	Primary users who access the LMS for learning activities	Access course materials, submit assignments, participate in quizzes, view grades, receive announcements, and monitor deadlines through the calendar system
Lecturers	Manage and deliver course content through the LMS	Upload lecture notes, create assignments and quizzes, monitor student participation, manage announcements, and provide grades and feedback
Administrators	Manage the overall LMS operation and user accounts	Maintain system configuration, manage users and courses, ensure system security, and monitor overall LMS performance
IT Support Staff	Provide technical maintenance and system support	Ensure system availability, solve technical issues, maintain server performance, and support users facing system problems
Faculty Management	Monitor academic activities and course performance	Access reports, monitor teaching and learning progress, and ensure effective use of the LMS within the university

The stakeholders identified have varying roles and expectations for the AIU Moodle LMS. The students are mainly concerned with accessing learning resources and fulfilling learning activities, and lecturers are mainly concerned with controlling the course content and monitoring the students' learning activities. IT support personnel and administrators ensure functionality, security, and reliability of the system. In addition, the system is a vital tool for faculty management in tracking academic performance and assisting institutional learning activities. Thus, it is significant to know the requirement of the stakeholders in order to determine the right functional and non-functional requirements of the system.

3. Requirements Elicitation and Documentation

Requirements elicitation is the activity of discovering, analyzing, and recording the requirements and expectations of the stakeholders of a software system. Requirements engineering is a way to assure system functionalities are in line with user needs and organizational goals, according to Sommerville (2016). The need for AIU Moodle LMS was identified in this report by observing the interface of the system, analysing the stakeholders and analysing the main features available in the platform.

The requirements identified are broken down into two types: Functional Requirements (FRs) and Non-Functional Requirements (NFRs). The functional requirements outline the services and functions that the system offers, whereas the non-functional requirements outline the quality attributes and operational constraints of the system (Pressman & Maxim, 2020).

3.1 Functional Requirements

Functional requirements describe the specific functions and services that the AIU Moodle LMS must provide for its users.

Req ID	Functional Requirement	Description
FR1	User Login	The system shall allow students, lecturers, and administrators to log into the AIU Moodle LMS using authorized credentials.
FR2	Course Access	The system shall allow students to access enrolled courses through the dashboard interface.
FR3	Upload Learning Materials	The system shall allow lecturers to upload lecture notes and course materials for students.
FR4	Assignment Submission	The system shall allow students to submit assignments electronically through the LMS submission feature.
FR5	Calendar Deadlines and	The system shall allow users to view assignment deadlines and academic schedules through the calendar feature.
FR6	Grade Viewing	The system shall allow students to view grades and feedback for completed assessments and assignments.
FR7	View Participant	The system shall allow lecturers and students to access participant information within enrolled courses.

Req ID	Functional Requirement	Description
FR8	Announcements and Notifications	The system shall display announcements and notifications related to courses and academic activities.

3.2 Non-Functional Requirements

Non-functional requirements describe the quality standards and operational characteristics required by the system.

Req ID	Non-Functional Requirement	Description
NFR1	Security	The system shall authenticate users securely to prevent unauthorized access to academic information and user accounts.
NFR2	Performance	The system shall load dashboard pages and course materials within 3 seconds under normal operating conditions.
NFR3	Availability	The system shall be available 24 hours a day to support online learning activities for students and lecturers.
NFR4	Usability	The system interface shall be user-friendly and easy to navigate for all users of the AIU Moodle LMS.

The identified functional and non-functional requirements reflect the main academic and administrative activities supported by the AIU Moodle LMS. These requirements were derived from stakeholder needs and observations of the system functionalities available in the platform. Proper documentation of these requirements is important to ensure consistency between stakeholder expectations and system operations during requirements engineering activities.

4. Use Case Modelling

Use case modelling is a Requirements Engineering technique used to describe the interactions between users and a system in order to achieve specific goals. Use case diagrams help visualize system functionalities, actors, and user interactions in a simple and structured form. According to Sommerville (2016), use case modelling is useful for understanding system behaviour from the users' perspective and identifying how stakeholders interact with the system functions.

In the AIU Moodle LMS, several actors interact with the system to perform academic and administrative activities. The main actors identified for this report are Students,

Lecturers, and Administrators. Based on the identified functional requirements, use cases were developed to represent the major activities supported by the system.

4.1 Use Case Diagram

The use case diagram of the AIU Moodle LMS shows the interactions among the main actors and the functions of the system. The diagram shows major academic and administrative activities like login, access to courses, submission of assignments, viewing of grades, upload of learning materials, access to participants, administration of announcements, supervision of the system and administration of users.

Main Actors

- Student
- Lecturer
- Administrator

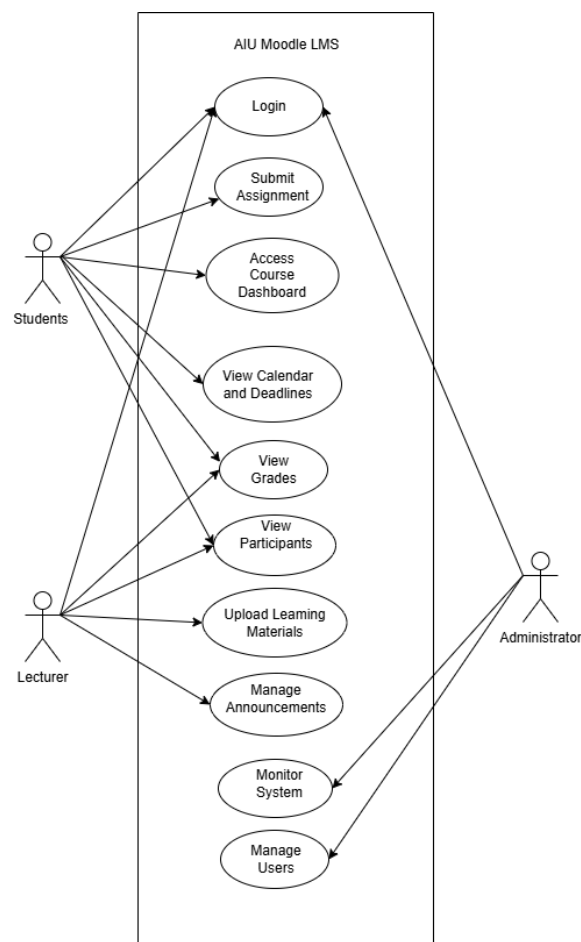


Figure 1: Use Case Diagram of AIU Moodle LMS

4.2 Use Case Descriptions

Use Case 1

Field	Description
Use Case ID	UC1
Use Case Name	Login
Primary Actor	Student, Lecturer, Administrator
Description	Users log into the AIU Moodle LMS using authorized credentials.
Preconditions	The user must have a registered LMS account.
Main Flow	1. The user enters a username and password. 2. The system verifies the credentials. 3. The system grants access to the dashboard.
Postconditions	The user successfully accesses the LMS dashboard.

Use Case 2

Field	Description
Use Case ID	UC2
Use Case Name	Submit Assignment
Primary Actor	Student
Description	Students submit assignments electronically through the LMS submission feature.
Preconditions	The student must be logged into the system and enrolled in the course.
Main Flow	1. The student opens the assignment page. 2. The student uploads the assignment file. 3. The student submits the assignment.
Postconditions	The assignment submission is recorded successfully in the system.

Use Case 3

Field	Description
Use Case ID	UC3
Use Case Name	Upload Learning Materials
Primary Actor	Lecturer
Description	Lecturers upload lecture notes and learning materials for students.
Preconditions	The lecturer must be logged into the LMS.
Main Flow	1. The lecturer accesses the course page. 2. The lecturer uploads learning materials. 3. The system stores the uploaded files.
Postconditions	The uploaded learning materials become accessible to students.

Use Case 4

Field	Description
Use Case ID	UC4
Use Case Name	View Grades
Primary Actor	Student, Lecturer
Description	Students and lecturers view grades and feedback for completed assessments and assignments.
Preconditions	The user must be logged into the LMS.
Main Flow	1. The user opens the grades section. 2. The system displays grades and feedback information.
Postconditions	The user successfully views assessment results and feedback.

Use Case 5

Field	Description
Use Case ID	UC5

Field	Description
Use Case Name	Manage Announcements
Primary Actor	Lecturer
Description	Lecturers create and manage course announcements for students.
Preconditions	The lecturer must be logged into the LMS.
Main Flow	1. The lecturer opens the announcement section. 2. The lecturer creates or updates an announcement. 3. The system publishes the announcement.
Postconditions	Students receive updated course announcements successfully.

The created use case model is the most significant interactions between the stakeholders and the AIU Moodle LMS. The identified functional requirements were matched with the identified stakeholder need to develop the use cases in a consistent manner between system functionalities and user's expectation. These models can be used to give more insight into user behaviour with the system and are used for documenting and analysing the requirements of the system throughout the requirements engineering process.

5. Requirements Prioritisation (MoSCoW Technique)

Prioritisation of Requirements is a crucial task in the Requirements Engineering process as it contributes to the determination of the importance of each of the requirements for the system according to the requirements of the stakeholders and the needs of the system. Requirements are often prioritized using the MoSCoW technique which has four categories – Must Have, Should Have, Could Have, Won't Have. Prioritisation can be used to ensure that critical functionalities of the system are carried out first, and that less critical functionalities are taken up later, as suggested by Sommerville (2016).

The functional requirements of the AIU Moodle LMS were prioritised using MoSCoW technique according to the importance of the features for activities of teaching, learning and managing the system.

Req ID	Requirement	Priority	Justification
FR1	User Login	Must Have	Users must log into the system to access LMS functionalities securely.

Req ID	Requirement	Priority	Justification
FR2	Access Course Dashboard	Must Have	Students require access to enrolled courses and academic information.
FR3	Upload Learning Materials	Must Have	Lecturers need to provide lecture notes and learning resources for students.
FR4	Submit Assignment	Must Have	Assignment submission is one of the core academic activities in the LMS.
FR5	View Calendar and Deadlines	Should Have	Calendar features help students manage deadlines and schedules effectively.
FR6	View Grades	Must Have	Students and lecturers need access to academic performance and feedback information.
FR7	View Participants	Could Have	Viewing participants supports communication and course collaboration but is not a core learning function.
FR8	Manage Announcements	Should Have	Announcements improve communication between lecturers and students.

The prioritisation results indicate that the main activities related to the access of learning, submission of assignments, learning materials, login, and the management of grades are classified as “Must Have” activities since they are of critical importance to the operation of the AIU Moodle LMS. Announcements, calendar scheduling, etc. are considered “Should Have” as these features enhance communication and organization in the system. The participant viewing feature is considered “Could Have” as it is designed to help with collaboration, but is not essential to the core functionality of the LMS.

6. Traceability Matrix

A Requirements Engineering tool to set up links between system requirements and other development elements like the use cases, testing activities and system functionalities, is called traceability matrix. Pressman and Maxim (2020) state that requirements traceability aids in the proper attention to identified requirements and their implementation in the system. Traceability also contributes to the consistency, management of requirements and system validation in software development.

This report created the traceability matrix to associate functional requirements of the AIU Moodle LMS to use cases. This helps to ensure each identified requirement has associated system interactions and functions.

Req ID	Functional Requirement	Related Use Case ID(s)	Use Case Name(s)	Status
FR1	User Login	UC1	Login	Completed
FR2	Access Course Dashboard	UC1, UC2	Login, Access Course Dashboard	Completed
FR3	Upload Learning Materials	UC1, UC3	Login, Upload Learning Materials	Completed
FR4	Submit Assignment	UC1, UC2	Login, Submit Assignment	Completed
FR5	View Calendar and Deadlines	UC1, UC5	Login, View Calendar and Deadlines	Completed
FR6	View Grades	UC1, UC4	Login, View Grades	Completed
FR7	View Participants	UC1, UC6	Login, View Participants	Completed
FR8	Manage Announcements	UC1, UC5	Login, Manage Announcements	Completed

The traceability matrix shows that there is a connection between all functional requirements identified and the use cases identified in the Moodle AIU. This mapping ensures consistency between the needs of the stakeholders, their functional requirements, and what the system does. In addition, the traceability matrix can be used to improve the requirement management process by keeping it well aligned with no important requirement missing in the requirement engineering process.

7. Tool Usage Explanation

Requirements Engineering tools play an important role in supporting requirement analysis, system modelling, documentation, and project organization. The use of appropriate tools helps improve the accuracy, consistency, and management of system requirements throughout the software engineering process. In this report, several tools were used to support the analysis and documentation of the AIU Moodle LMS.

7.1 Draw.io

Draw.io was used to create the UML use case diagram for the AIU Moodle LMS. Draw.io is a user-friendly diagramming tool that provides various UML templates and diagram components. The tool was selected because it allows clear visualization of

system actors, use cases, and system interactions in a simple and organized manner.

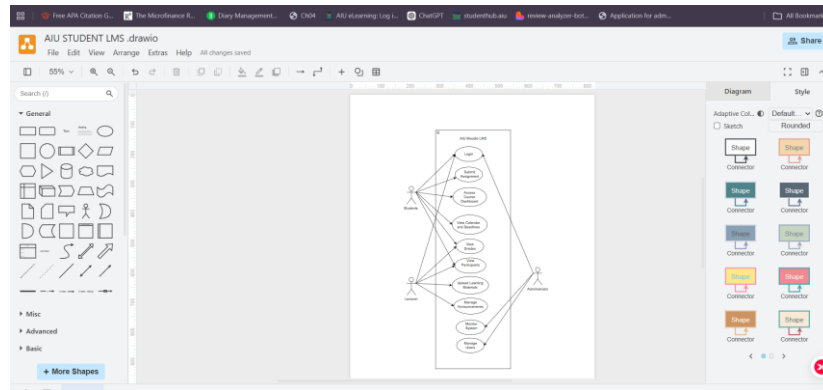


Figure 2: Use Case Diagram created using Draw.io

7.2 Wiki Technologies

Wiki technologies were used to support requirements engineering activities by organizing, documenting, and managing system-related information during the analysis of the AIU Moodle LMS. Wiki technologies support collaborative documentation and structured information management in requirements engineering processes. According to the lecture materials, wiki technologies are useful for organizing glossaries, documenting requirements, and supporting cooperation among stakeholders, especially in systems involving multiple users and system functionalities. Wiki tools help improve information sharing, requirement organization, and documentation consistency.

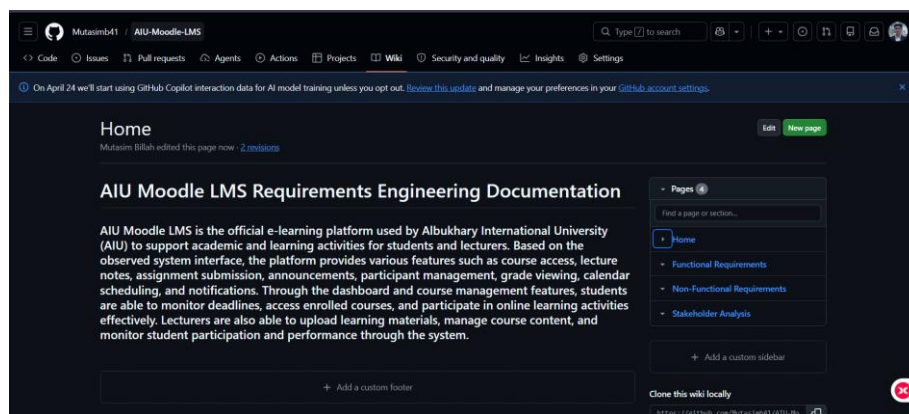


Figure 3: Wiki technologies used to support requirements engineering documentation and information management

7.3 Trello

Trello was used for organizing and managing system requirements and task progression during the report preparation process. Trello provides a simple project management interface using boards, lists, and cards. The tool helps organize requirements, tasks, and report sections systematically.

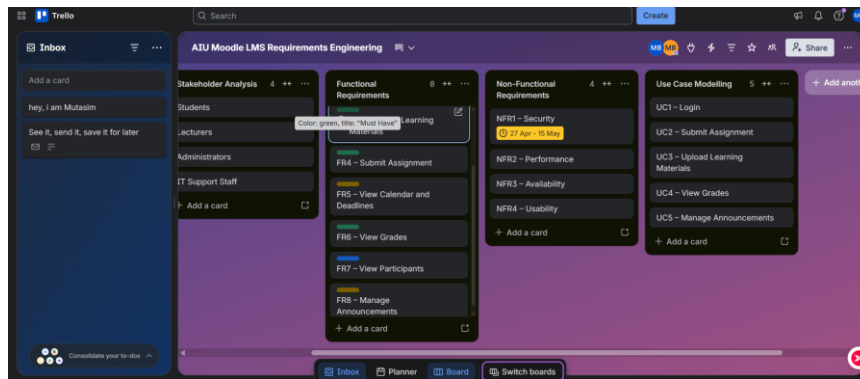


Figure 4: Trello board used for requirements organization

7.4 GitHub

GitHub was used for documentation management and version control during report preparation. GitHub supports file management, version tracking, and document organization. The tool helps maintain different versions of diagrams, report content, and requirement documentation systematically.

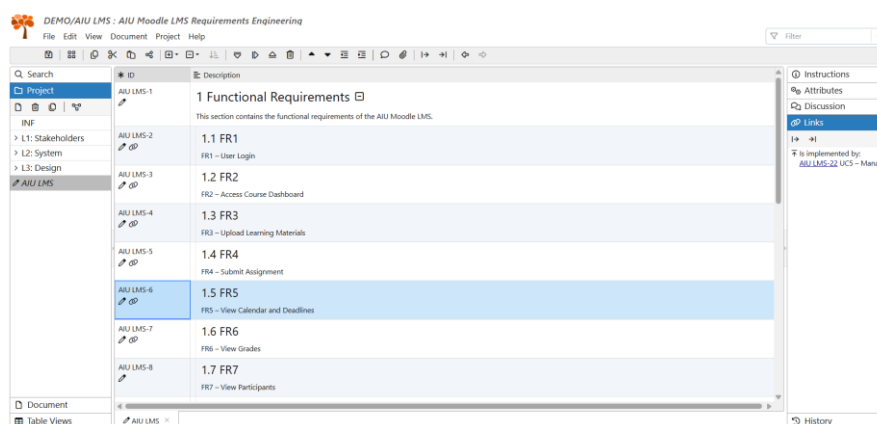
Figure 5 :GitHub repository used for documentation management

7.5 ReqView

ReqView was used to support requirement organization and traceability management. ReqView is a requirements management tool that helps organize functional and non-functional requirements and supports requirement traceability activities.

Output Produced

- Requirement organization
- Traceability management
- Requirement documentation



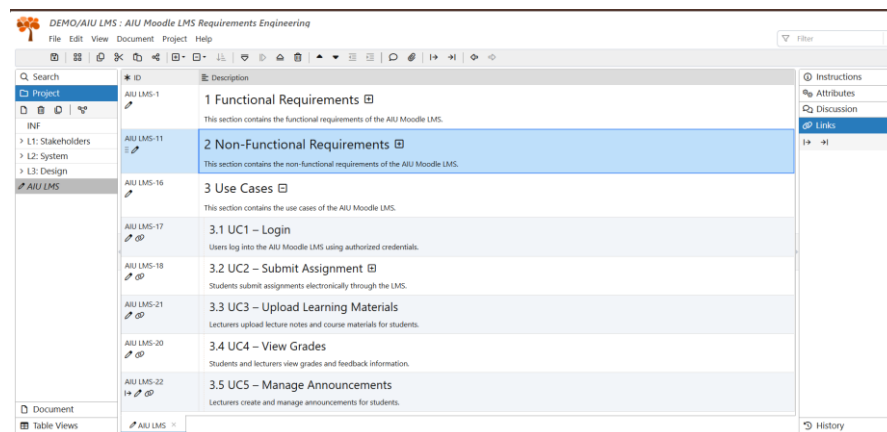


Figure 6: ReqView interface used for requirements management

8. Conclusion

The Requirements Engineering analysis conducted for the AIU Moodle LMS helped identify and document the major functional and non-functional requirements of the system. Through stakeholder analysis, requirement elicitation, use case modelling, prioritisation, and traceability activities, the report provided a structured understanding of how different users interact with the LMS and how the system supports academic and administrative activities within the university.

The identified functional requirements focused on important LMS features such as user login, course access, assignment submission, uploading learning materials, viewing grades, announcements, and calendar management. In addition, non-functional requirements such as security, performance, availability, and usability were identified to ensure that the system operates effectively and provides a reliable learning environment for users.

Use case modelling and the traceability matrix helped maintain consistency between stakeholder needs, functional requirements, and system interactions. The application of the MoSCoW prioritisation technique also helped classify system requirements according to their importance and contribution to the LMS operation.

Furthermore, the use of Requirements Engineering tools such as Draw.io, Trello, GitHub Wiki, and ReqView improved the organization, documentation, modelling, and management of requirements throughout the analysis process. These tools supported better visualization, traceability, and structured documentation of system requirements.

9. Reflection

Throughout the development of this Requirements Engineering report, several challenges were encountered during the analysis and documentation of the AIU Moodle LMS. One of the main challenges was identifying and organizing system requirements accurately based on the available LMS features and stakeholder

interactions. Since the system contains multiple academic and administrative functionalities, it was important to ensure that all functional and non-functional requirements were clearly defined and aligned with stakeholder needs.

Another challenge was maintaining consistency between the different sections of the report, especially between the functional requirements, use case modelling, MoSCoW prioritisation, and traceability matrix. Ensuring that requirement IDs, use case IDs, and system functionalities remained properly connected throughout the report required careful organization and review.

The use of Requirements Engineering tools helped address many of these challenges effectively. Draw.io assisted in creating clear and organized UML use case diagrams that improved understanding of system interactions and actor relationships. Trello helped organize tasks, requirements, and report progress systematically through a visual workflow structure. GitHub Wiki supported centralized documentation and structured organization of stakeholder information, requirements, and use cases. In addition, ReqView helped manage requirements and traceability relationships more efficiently by providing structured requirement organization and linking capabilities.

10. References

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